

System Information and Network Testing - Kali Linux

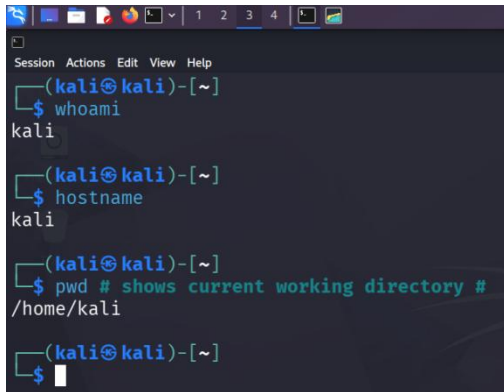
Submitted by

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Part 1

Step 1: System Information- Kali Linux Virtual Machine



```
(kali㉿kali)-[~]  
$ whoami  
kali  
  
(kali㉿kali)-[~]  
$ hostname  
kali  
  
(kali㉿kali)-[~]  
$ pwd # shows current working directory #  
/home/kali  
  
(kali㉿kali)-[~]  
$
```

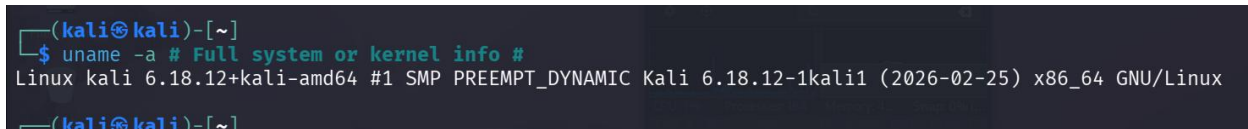
The username is kali.

Machine host name is kali

Current working directory is /home/kali

Step 2

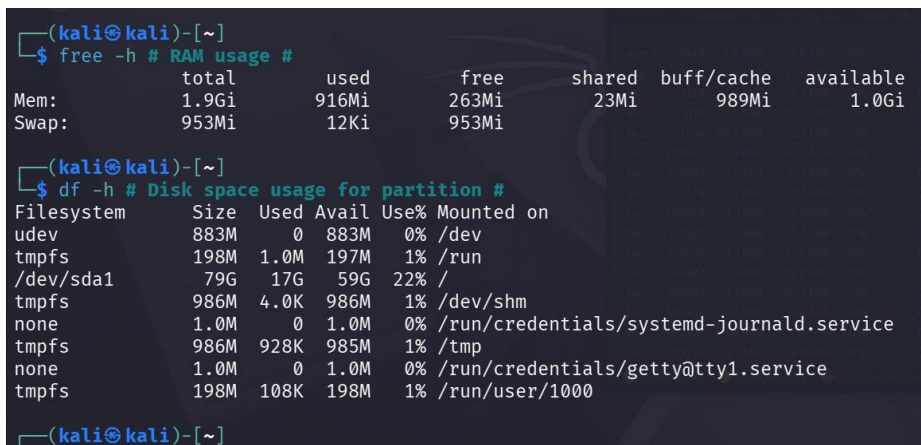
- Operating Systems and Hardware Information.



```
(kali㉿kali)-[~]  
$ uname -a # Full system or kernel info #  
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux  
  
(kali㉿kali)-[~]
```

The version of the linux kernel of this kali machine is 6.18.12 -amd64

- RAM usage and Disk Space Available



```
(kali㉿kali)-[~]  
$ free -h # RAM usage #  


|       | total | used  | free  | shared | buff/cache | available |
|-------|-------|-------|-------|--------|------------|-----------|
| Mem:  | 1.9Gi | 916Mi | 263Mi | 23Mi   | 989Mi      | 1.0Gi     |
| Swap: | 953Mi | 12Ki  | 953Mi |        |            |           |

  
(kali㉿kali)-[~]  
$ df -h # Disk space usage for partition #  


| Filesystem | Size | Used | Avail | Use% | Mounted on                                |
|------------|------|------|-------|------|-------------------------------------------|
| udev       | 883M | 0    | 883M  | 0%   | /dev                                      |
| tmpfs      | 198M | 1.0M | 197M  | 1%   | /run                                      |
| /dev/sda1  | 79G  | 17G  | 59G   | 22%  | /                                         |
| tmpfs      | 986M | 4.0K | 986M  | 1%   | /dev/shm                                  |
| none       | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-journald.service |
| tmpfs      | 986M | 928K | 985M  | 1%   | /tmp                                      |
| none       | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/getty@tty1.service       |
| tmpfs      | 198M | 108K | 198M  | 1%   | /run/user/1000                            |

  
(kali㉿kali)-[~]
```

The total RAM of this virtual machine is 1.9 gigabytes.

The disk space available in the main partition is 59 gigabytes.

Part 2

Step 1: Basic network connectivity test

```
(kali㉿kali)-[~]  
$ ping -c 2 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=47.4 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=26.1 ms  
  
— 8.8.8.8 ping statistics —  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 26.081/36.723/47.366/10.642 ms  
  
(kali㉿kali)-[~]
```

Average round-trip time is **36.723 ms**

Packet loss? No packet loss

Step 2

- Internet connectivity and DNS resolution

```
(kali㉿kali)-[~]  
$ ping -c 4 google.com  
PING google.com (142.251.216.110) 56(84) bytes of data.  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=1 ttl=117 time=59.5 ms  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=2 ttl=117 time=255 ms  
64 bytes from lclsh-ad-in-f14.1e100.net (142.251.216.110): icmp_seq=4 ttl=117 time=343 ms  
  
— google.com ping statistics —  
4 packets transmitted, 3 received, 25% packet loss, time 3009ms  
rtt min/avg/max/mdev = 59.481/219.056/342.822/118.412 ms  
  
(kali㉿kali)-[~]  
$
```

Resolved IP address of Google is **142.251.216.110**

Round Trip Time (rtt)

The round trip time (rtt) in pinging google.com is **219.056 ms** and it is different from, and way more than the rtt in pinging 8.8.8.8 directly.

The possible reasons are below:

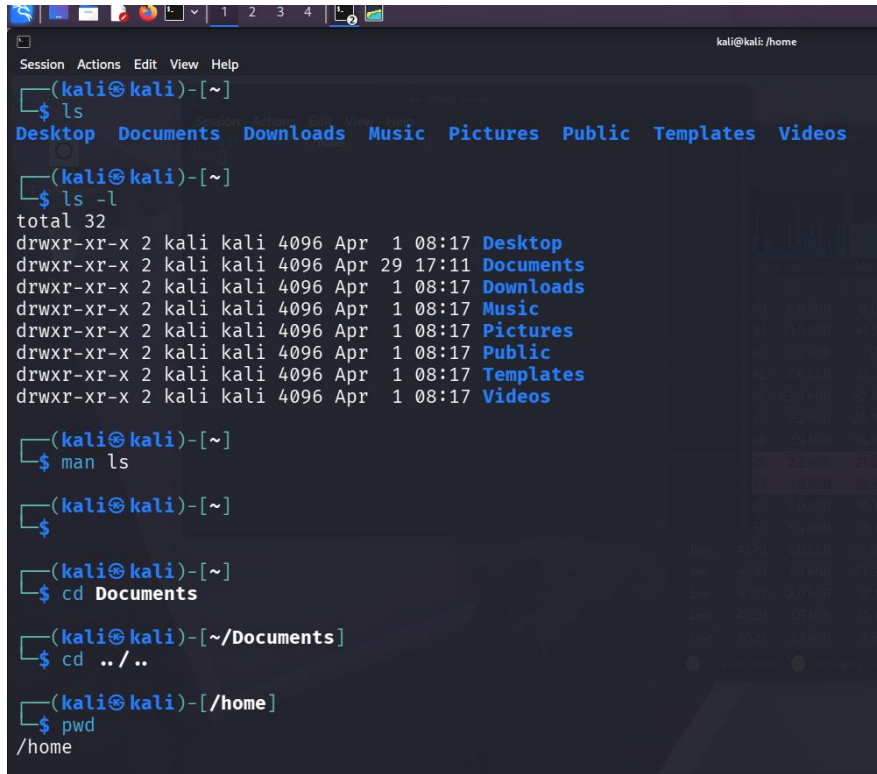
- **Different servers:** 8.8.8.8 is google public DNS server while google.com resolves to a google web server/CDN IP which is not 8.8.8.8. So, as the packets are sent to different servers, they will take different paths and will have different latencies.
- **DNS Lookup Time:** The first ping to Google.com will require a DNS resolution of Google.com IP address. This will have a DNS overhead and add a few ms on the first packet only, while 8.8.8.8 will skip that step as it is already an IP.
- **Packet count:** the packet count `-c`, affects RTT. As I pinged 8.8.8.8 with `-c 2`, the first packet tends to be slower due to ARP request and the second packet skips ARP because it has been caught. Meanwhile a ping on google.com with `-c 4` tend to result in 1 slow packet and 3 fast packets, making its average RTT slower.

What is `-c 4` ?

`-c 4` means 'count is 4'. It prompts ping to send 4 ICMP echo request packets then stop.

Other Deliverables:

- Listing and changing directories, and exploring man page.



```
(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos

(kali㉿kali)-[~]
└─$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Desktop
drwxr-xr-x 2 kali kali 4096 Apr 29 17:11 Documents
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Downloads
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Music
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Pictures
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Public
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Templates
drwxr-xr-x 2 kali kali 4096 Apr  1 08:17 Videos

(kali㉿kali)-[~]
└─$ man ls

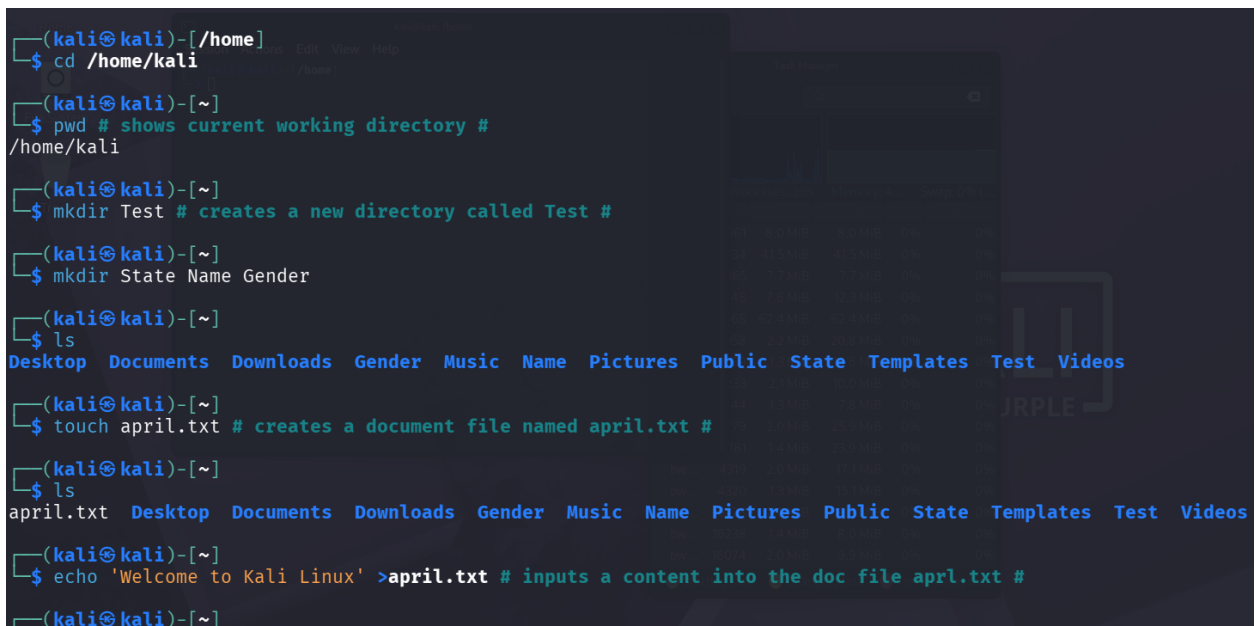
(kali㉿kali)-[~]
└─$

(kali㉿kali)-[~]
└─$ cd Documents

(kali㉿kali)-[~/Documents]
└─$ cd ../../

(kali㉿kali)-[/home]
└─$ pwd
/home
```

- Creating files and directories, and inputting content into a file.



```
(kali㉿kali)-[/home]
└─$ cd /home/kali

(kali㉿kali)-[~]
└─$ pwd # shows current working directory #
/home/kali

(kali㉿kali)-[~]
└─$ mkdir Test # creates a new directory called Test #

(kali㉿kali)-[~]
└─$ mkdir State Name Gender

(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Gender  Music  Name  Pictures  Public  State  Templates  Test  Videos

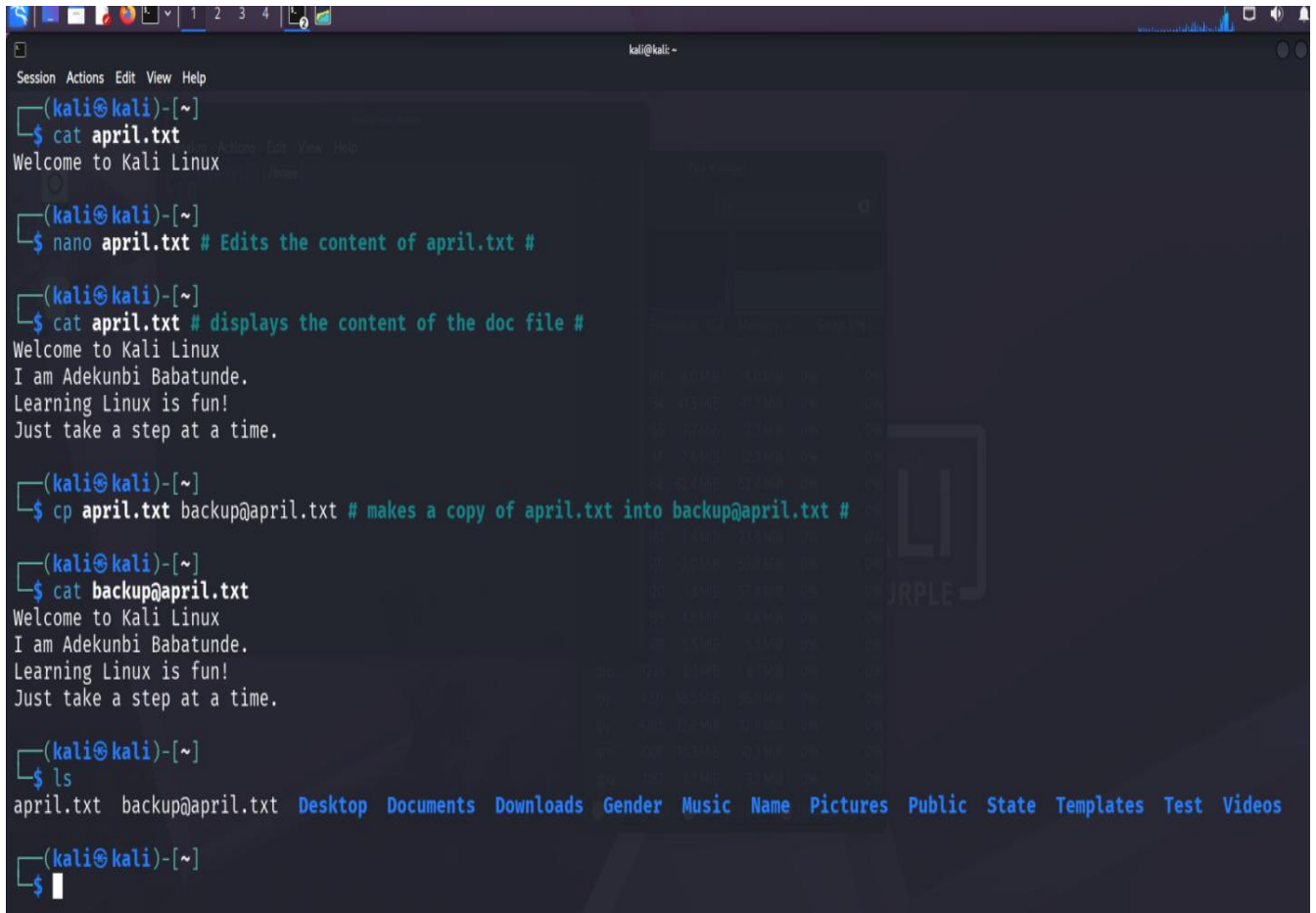
(kali㉿kali)-[~]
└─$ touch april.txt # creates a document file named april.txt #

(kali㉿kali)-[~]
└─$ ls
april.txt  Desktop  Documents  Downloads  Gender  Music  Name  Pictures  Public  State  Templates  Test  Videos

(kali㉿kali)-[~]
└─$ echo 'Welcome to Kali Linux' >april.txt # inputs a content into the doc file aprl.txt #

(kali㉿kali)-[~]
```

Displaying, copying and editing file content.

A terminal window titled 'kali@kali' showing a series of commands and their outputs. The user first runs 'cat april.txt', which displays a welcome message and a personal introduction. Then, they use 'nano april.txt' to edit the file. After saving, they run 'cat april.txt' again to confirm the changes. Next, they use 'cp april.txt backup@april.txt' to create a backup. Finally, they run 'cat backup@april.txt' to verify the backup content. The terminal also shows the output of 'ls' at the bottom, listing files in the current directory.

```
(kali@kali)-[~]
$ cat april.txt
Welcome to Kali Linux

(kali@kali)-[~]
$ nano april.txt # Edits the content of april.txt #

(kali@kali)-[~]
$ cat april.txt # displays the content of the doc file #
Welcome to Kali Linux
I am Adekunbi Babatunde.
Learning Linux is fun!
Just take a step at a time.

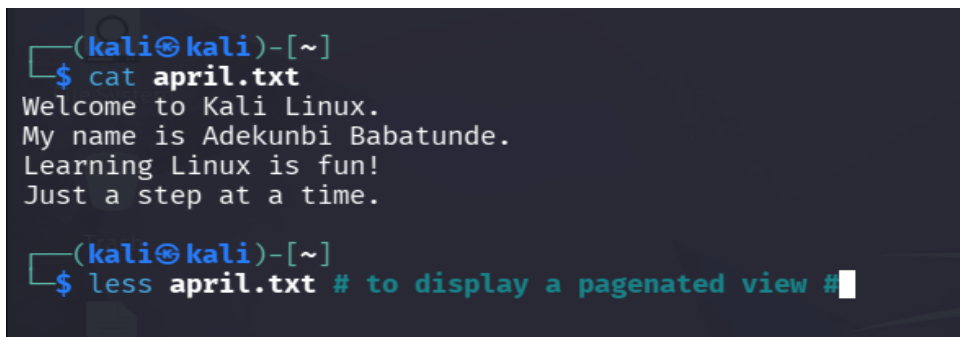
(kali@kali)-[~]
$ cp april.txt backup@april.txt # makes a copy of april.txt into backup@april.txt #

(kali@kali)-[~]
$ cat backup@april.txt
Welcome to Kali Linux
I am Adekunbi Babatunde.
Learning Linux is fun!
Just take a step at a time.

(kali@kali)-[~]
$ ls
april.txt  backup@april.txt  Desktop  Documents  Downloads  Gender  Music  Name  Pictures  Public  State  Templates  Test  Videos

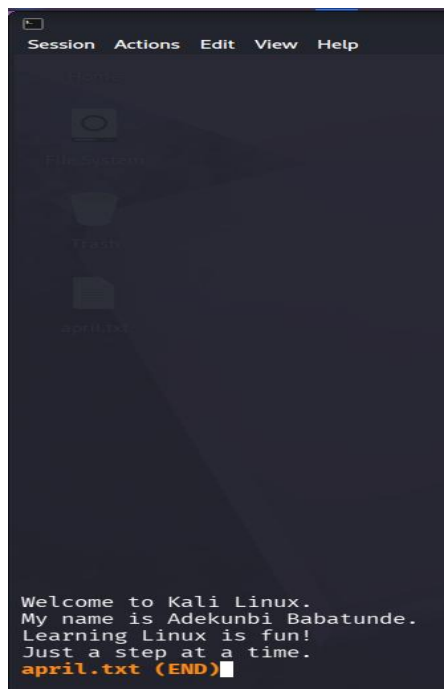
(kali@kali)-[~]
$
```

- Exploring less command

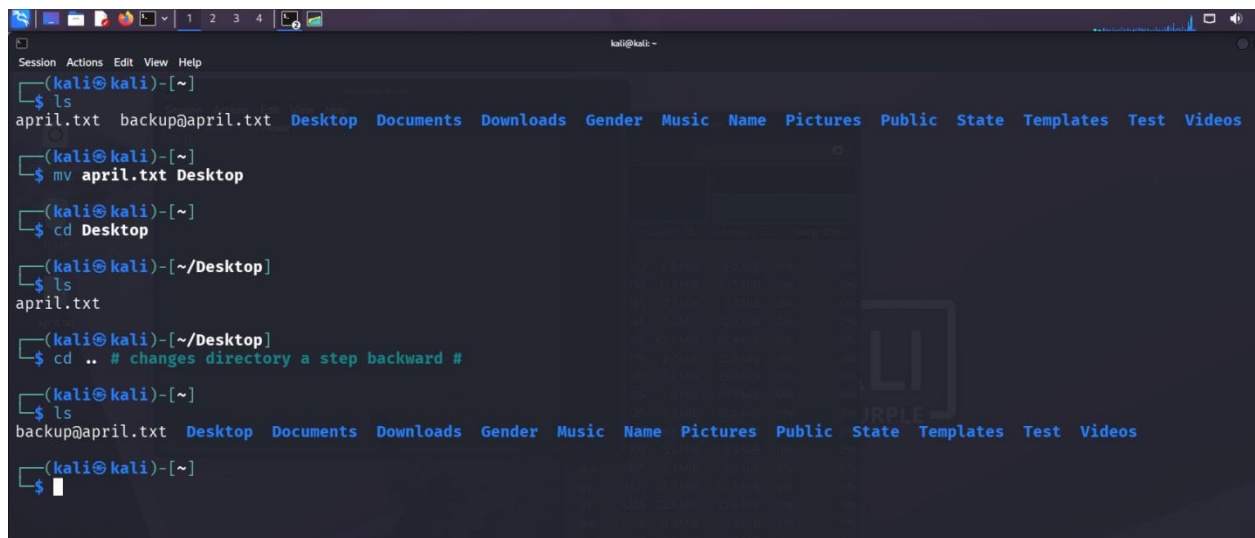
A terminal window titled '(kali@kali)-[~]' showing the user running 'cat april.txt' and then 'less april.txt'. The output of 'cat' shows the same welcome message and introduction as the previous screenshot. The 'less' command is entered at the end of the terminal line, ready to be executed.

```
(kali@kali)-[~]
$ cat april.txt
Welcome to Kali Linux.
My name is Adekunbi Babatunde.
Learning Linux is fun!
Just a step at a time.

(kali@kali)-[~]
$ less april.txt # to display a pagenated view #
```



- Changing and moving directories.



- Removing files and directories.

```

kali@kali ~
Session Actions Edit View Help

(kali@kali)-[~]
$ ls
backup@april.txt Desktop Documents Downloads Gender Music Name Pictures Public State Templates Test Videos

(kali@kali)-[~]
$ rm backup@april.txt # removes backup@april.txt file #

(kali@kali)-[~]
$ rmdir Name Test # removes Name and Test directories #

(kali@kali)-[~]
$ ls
Desktop Documents Downloads Gender Music Pictures Public State Templates Videos

(kali@kali)-[~]
$

```

- Installing updates and advance package tools in Kali

```

(kali@kali)-[~]
$ sudo apt update
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/non-free amd64 Packages [189 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [908 kB]

```

```

(kali@kali)-[~]
$ sudo apt install curl # Installs curl #
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
base58 python3-browser-cookie3 python3-pyfiglet
binutils-mingw-w64-base python3-cbor python3-pyotp
binutils-mingw-w64-i686 python3-celery python3-pyqrcode
binutils-mingw-w64-x86-64 python3-cli-helpers python3-pyramid
debugedit python3-click-didyoumean python3-pyshodan
dnsmap python3-click-repl python3-qasync
dsniff python3-cmd2 python3-qrcode
ettercap-common python3-configobj python3-serial-asyncio
ettercap-graphical python3-cpuinfo python3-sgmllib3k
faraday-agent-dispatcher python3-cvss python3-sh
figlet python3-ecdsa python3-shtab
finger python3-elastic-transport python3-simple-rest-client
gcc-mingw-w64-base python3-elasticsearch python3-simplekv
gcc-mingw-w64-i686-win32 python3-ephem python3-slugify
gcc-mingw-w64-i686-win32-runtime python3-faraday-agent-parameters-types python3-smmmap
gcc-mingw-w64-x86-64-win32 python3-faraday-plugins python3-snappy
gcc-mingw-w64-x86-64-win32-runtime python3-feedparser python3-sniffio
gir1.2-vte-2.91 python3-filedepot python3-spinners
greenbone-feed-sync python3-filteralchemy python3-sqlalchemy-schemadisplay
gvmd python3-flask-celery-helper python3-standard-imghdr
gvmd-common python3-flask-classful python3-status
imagemagick python3-flask-kvsession python3-syslog-rfc5424-formatter
imagemagick-7.q16 python3-flask-limiter python3-tables
libapache2-mod-php python3-flask-login python3-tables-lib

```



```
(kali㉿kali)-[~]
└─$ curl --version # confirms curl installation #
curl 8.19.0 (x86_64-pc-linux-gnu) libcurl/8.19.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2 libssh
2/1.11.1 nghttp2/1.68.0 ngtcp2/1.21.0 nghttp3/1.12.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-03-11, security patched: 8.19.0-3
```

```
(kali㉿kali)-[~]
└─$ sudo apt remove curl # removes curl #
The following packages were automatically installed and are no longer required:
base58 python3-browser-cookie3 python3-pyfiglet
binutils-mingw-w64-base python3-cbor python3-pyotp
binutils-mingw-w64-i686 python3-celery python3-pyqrcode
binutils-mingw-w64-x86-64 python3-cli-helpers python3-pyramid
debugedit python3-click-didyoumean python3-pyshodan
dnsmasq python3-click-repl python3-qasync
dsniff python3-cmd2 python3-qrcode
ettercap-common python3-configobj python3-serial-asyncio
ettercap-graphical python3-cpuinfo python3-sgmllib3k
faraday-agent-dispatcher python3-cvss python3-sh
figlet python3-ecdsa python3-shtab
finger python3-elastic-transport python3-simple-rest-client
gcc-mingw-w64-base python3-elasticsearch python3-simplekv
gcc-mingw-w64-i686-win32 python3-ephem python3-slugify
gcc-mingw-w64-i686-win32-runtime python3-faraday-agent-parameters-types python3-smmmap
gcc-mingw-w64-x86-64-win32 python3-faraday-plugins python3-snappy
gcc-mingw-w64-x86-64-win32-runtime python3-feedparser python3-sniffio
gir1.2-vte-2.91 python3-filedepot python3-spinners
greenbone-feed-sync python3-filteralchemy python3-sqlalchemy-schemadisplay
gvmd python3-flask-celery-helper python3-standard-imghdr
```

```
Use 'sudo apt autoremove' to remove them.
```

```
REMOVING:
curl

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1114
  Freed space: 513 kB

Continue? [Y/n] y
(Reading database... 392514 files and directories currently installed.)
Removing curl (8.19.0-3)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...
```

```
(kali㉿kali)-[~]
└─$
```

```
(kali㉿kali)-[~]
└─$ curl --version # confirms curl installation removal #
Command 'curl' not found, but can be installed with:
sudo apt install curl
Do you want to install it? (N/y)
```